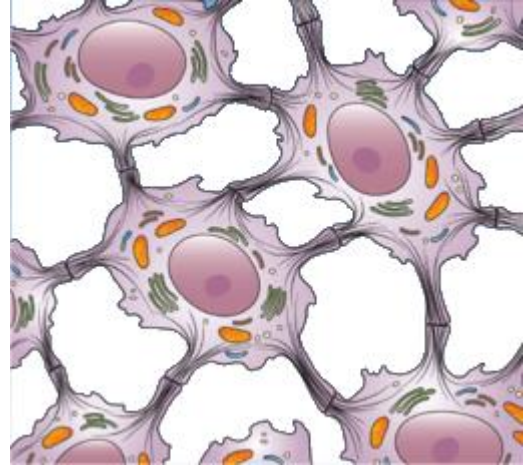


Name: _____

Why Are Cells So Small?

Introduction:

The smallest objects that can be seen by the human eye are about 0.1 mm long. Some single-celled organisms are just big enough that you could see them. A human egg is about 0.12 mm in diameter. The vast majority of cells in your body and cells that form bacteria are too small to see without a microscope. Usually, when a cell gets too large, it divides or stops growing. Cells remain small because all materials that are exchanged between the cell and its environment, like oxygen and glucose, must pass through the cell membrane. If materials cannot be exchanged efficiently, then the cell could die.



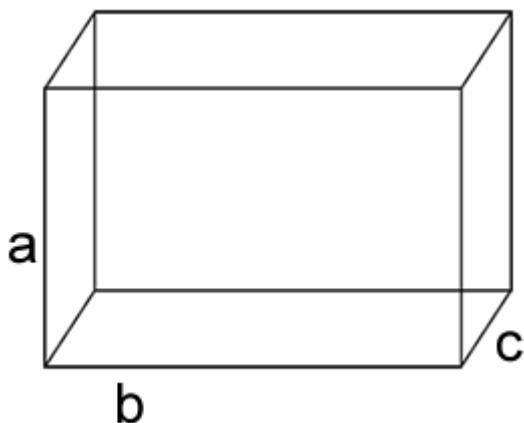
In this activity, you will explore how surface area and volume limit the size of cells.

Procedure

Your group will receive a box (tissue box, jewelry box, food box) and your task is to measure the surface of the box, and also measure the volume of the box.

1. Measure the surface area by measuring the area of each side of the box and adding those numbers. Show your work below!

$$\text{Surface area} = 2 \times a \times b + 2 \times b \times c + 2 \times a \times c$$



2. Measure the volume of the box.

$$\text{Volume} = a \times b \times c$$

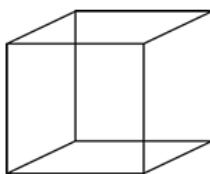
3. Calculate the Surface Area to Volume ratio.

$$\text{SA} / \text{Volume}$$

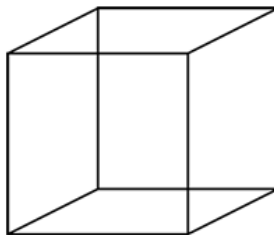
3. Compile Data - All of the boxes in the room are different sized. Fill out the chart below by collaborating with your classmates.

Surface Area	Volume	SA/Volume

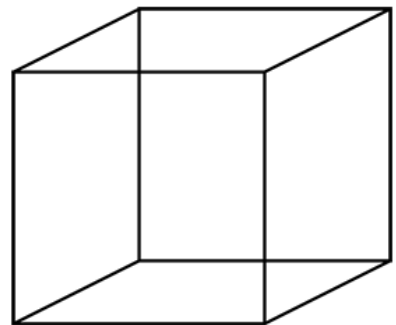
4. Measure the surface area and the volume of each of the cubes below. Show your work!



1 cm



2 cm



3 cm

	Box 1	Box 2	Box 3
Surface Area			
Volume			
SA/Ratio			

Analysis

1. Consider the data from the boxes you measured and the cubes. As the boxes (cubes) get larger, what happens to their surface area / volume ratio?

2. Explain how this ratio would relate to limits on cell size. Be careful and thoughtful with your response. (You may need to reread the introduction of this activity.)

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Cell Image: https://commons.wikimedia.org/wiki/File:Desmosome_Cell_Junction.png

Teacher's note: I collect boxes from food, like mac and cheese, and any other box-shaped items. You can also make boxes out of construction paper using templates, like [this one](#).

HS-LS2-1- Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells.

HS-LS1-2. Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.